

Project ATLAS

Explained Like You're Five — With the Real Math Included

A plain-language walkthrough of why ATLAS exists, what problem it solves, and exactly how it works underneath — for readers who want both the bedtime-story version and the textbook version, side by side.

*Draft explainer document — not the investment memo or the technical spec.
Prepared July 2026.*

Table of Contents

1. Why Do We Even Need This?
2. What Exactly Is Broken? (The Problem, Precisely)
3. The Big Idea: Data Has a Shape
4. Talking Without Telling: The TIP Handshake
5. Visiting Without Taking Anything Home: S-MCP
6. A Tiny Worked Example, With Real Numbers
7. What's Actually True Today (The Honest Part)

1. Why Do We Even Need This?

Today's AI systems answer questions by turning documents into long lists of numbers (called embeddings) and then finding the list that looks most similar to your question. This works fine for simple lookups, but it breaks down in three specific ways:

- It can't follow a chain of clues across several documents — it only compares one thing to one other thing at a time.
- It can't notice when something is structurally weird (like fraud), because weirdness is about how a thing relates to its neighbors, not just its raw distance from a query.
- It has no safe way for two separate companies (say, two banks) to reason over each other's data without physically sending the data over — which is illegal or forbidden in a lot of industries.

LIKE YOU'RE FIVE

Imagine you and your friend each have a box of LEGO. You both secretly built a castle, and you want to know if your castles have a matching secret tower somewhere — but neither of you is allowed to show the other your whole box. Right now, there's no good way to check. ATLAS is a way to compare the “shapes” of your castles without ever opening the box.

ATLAS's answer is to stop treating data as flat text and start treating it as a geometric shape — a manifold — that can be compared, traversed, and reasoned over the way you'd reason over a map, not a word search.

2. What Exactly Is Broken? (The Problem, Precisely)

Standard retrieval systems (Retrieval-Augmented Generation, or RAG) embed each document as a vector, then rank documents by cosine similarity to the query vector.

THE REAL MATH — Cosine Similarity

$$\cos(\theta) = (A \cdot B) / (||A|| * ||B||)$$

where A = query vector, B = document vector,

$A \cdot B$ = dot product, $||A||$ = vector length

This measures only the angle between two single points. It has no term for a third, fourth, or fifth document, and no term for how a group of documents sits relative to each other. That is the structural ceiling — no amount of extra data or a better embedding model changes what this formula is capable of expressing.

LIKE YOU'RE FIVE

It's like judging how close two kids are standing by measuring the angle between them and you — but never noticing that they're actually holding hands, or standing in a circle with three other kids. The angle trick only ever looks at two points at a time.

Three concrete failure modes follow directly from this:

- **Multi-hop questions** — an answer that requires chaining facts across records has no natural representation in a purely pointwise scoring system.
- **Structural anomalies** — fraud and defects are often unusual in their relationship to their neighborhood, not in raw distance from a query.
- **Cross-organization reasoning** — there is no mechanism for two organizations to jointly reason over data without centralizing it, which regulated industries cannot do.

3. The Big Idea: Data Has a Shape

ATLAS's first pillar, Manifold-Core, builds an actual geometric structure out of a dataset instead of a flat list of vectors. It does this in five steps:

- 1 Encode records into embeddings (same starting point as ordinary RAG).
- 2 Build an approximate k-nearest-neighbor graph connecting each point to its closest neighbors.
- 3 Grow a Vietoris–Rips filtration — connect points as an allowed distance is slowly increased.
- 4 Track which shapes (clusters, loops, voids) survive as that distance grows, versus which vanish quickly.
- 5 Answer a query by dropping it into this shape and letting it settle, like a marble rolling into a groove.

LIKE YOU'RE FIVE

Picture connect-the-dots, but the dots are ideas instead of stars. If you slowly stretch a rubber band bigger and bigger around a handful of dots, some dots snap together fast (they're close friends), and some stay lonely for a long time (they're outsiders). A loop of dots that stays a loop for a long time before filling in is probably an important shape — not an accident. ATLAS remembers which shapes were 'real' and which were just noise.

THE REAL MATH — Vietoris–Rips Complex and Persistence

$VR_e(X) = \{ \sigma \text{ (subset of) } X : d(x_i, x_j) \leq e \text{ for all } x_i, x_j \text{ in } \sigma \}$

As e (epsilon) increases from 0 to infinity, simplices are added one by one.

Each topological feature (a component, loop, or void) is 'born' at some threshold b and 'dies' (gets filled in / merges) at some threshold d .

$\text{persistence}(\text{feature}) = d - b$

A large persistence value ($d - b$) means the feature survived across a wide range of scales — treated as real structure. A small value means it appeared and vanished almost immediately — treated as noise. This single number is what lets the system separate signal from noise mathematically, rather than by a hand-tuned threshold.

Retrieval then isn't 'find the nearest point.' A query is inserted into the complex, and a potential field is defined using local point density plus nearby persistence weights. The query traverses that field — conceptually similar to gradient descent — so its answer can pass through several related regions in sequence. That is the mechanism that gives multi-hop reasoning a native representation.

WHERE THIS GETS HARD

Exact Vietoris–Rips persistence computation scales combinatorially in the number of simplices as dataset size and dimension grow. This is the single biggest open engineering problem in the whole system. Mitigations under investigation — sparse/witness-complex approximations, capping homology at low dimensions, and incremental updates — are reasonable, literature-backed ideas. None of them has yet been proven sufficient at production scale; that is exactly what the first implementation phase is designed to measure.

4. Talking Without Telling: The TIP Handshake

TIP (Topology Interchange Protocol) lets two organizations discover that they have matching structure in their data — without either one sending the other any raw records. It works on a *local* neighborhood around one specific query, not the whole dataset (an earlier, looser description of this protocol compared whole-dataset summaries; that version doesn't work, because a global summary can't say *where* an overlap is, only *that* the two datasets are shaped similarly overall).

LIKE YOU'RE FIVE

It's a secret handshake. You don't describe your whole day to your friend to check if you both like the same secret snack — you just do a special handshake, and if it matches, you both know without saying the snack's name out loud.

The four real steps:

- 1 Restrict the topology computation to the k-nearest neighborhood of the query point only.
- 2 Convert that local persistence diagram into a fixed-length vector using a persistence landscape transform.
- 3 Hash that vector using locality-sensitive hashing (random hyperplane projection).
- 4 Exchange only the hash codes. A collision resolves to a locally-scoped region id, meaningless outside the org that generated it.

THE REAL MATH — Persistence Landscape + Locality-Sensitive Hashing

Persistence landscape (turns a diagram into a comparable vector):

$\lambda_k(t) = k\text{-th largest value of } \min(t - b_i, d_i - t) \text{ over all birth/death pairs } (b_i, d_i), \text{ for a grid of } t \text{ values}$

Random-hyperplane LSH (turns a vector into a short, one-way hash):

$h_r(v) = \text{sign}(v \cdot r)$, r drawn randomly from a unit sphere

$\Pr[h_r(v_1) = h_r(v_2)] = 1 - \theta(v_1, v_2) / \pi$

$\theta(v_1, v_2)$ is the angle between the two vectors. The closer two organizations' local shapes are (small angle), the higher the chance their hashes collide. The vector itself cannot be recovered from the hash — only a match/no-match signal crosses the network.

SAID PLAINLY

This is probabilistic matching, not a cryptographic proof. False positives (coincidental collisions) and false negatives (near-matches that don't collide) are expected outcomes of the design and must be measured, not assumed away. A collision also leaks a small amount of information — the fact that a structural match exists — so 'zero-knowledge' is not the right term for this; 'privacy-preserving approximate matching' is. Repeated querying over time has no proven leakage bound yet; rate-limiting plus differential-privacy noise in the hashing step is the planned mitigation, not a solved problem.

5. Visiting Without Taking Anything Home: S-MCP

Once TIP finds a matching region, someone still needs to actually compute an answer using the data sitting at that region — without that data ever leaving its home organization. Sovereign-MCP (S-MCP) solves this by sending the *computation* to the data, instead of the data to the computation.

LIKE YOU'RE FIVE

Instead of asking your friend to mail you their favorite toy so you can check something about it, you mail your friend a tiny, harmless robot with one very specific, simple job. The robot goes in, looks only at the one toy it's allowed to see, writes down a short note, and comes straight back — it can't wander around the house, and it doesn't remember anything once it's done.

The execution flow:

- 1 A small, task-distilled inference module (not a general model) is trained offline for one narrow sub-question.
- 2 It's compiled to WebAssembly (wasm32-wasi), quantized to int8, and signed.
- 3 The host verifies the signature and runs it in a sandbox with a fixed memory ceiling, a fixed compute-step budget, and no filesystem or network access — only a narrow, read-only view of the matched region.
- 4 It returns a small result vector (at most 256 bytes).
- 5 The sandbox is torn down completely; no state persists.

There isn't new math here so much as a discipline: the module has already been told exactly where to look and what narrow question to ask, by TIP, before it is ever invoked — so it needs accuracy on one distilled task, not general reasoning.

6. A Tiny Worked Example, With Real Numbers

Two banks each suspect a shell company. Bank A and Bank B each hold a cluster of wire transfers around a suspicious account. Here's a simplified run-through with toy numbers, following the same steps used in the real design.

Step 1 — Local persistence.

Bank A's cluster produces one persistent loop with birth $b = 0.10$ and death $d = 0.85$. Persistence $= 0.85 - 0.10 = 0.75$ — a strong, real feature (not noise).

Bank B's cluster, around its own suspicious account, produces a similarly-shaped loop with birth $b = 0.12$ and death $d = 0.9$, persistence $= 0.78$.

Step 2 — Landscape vectors.

Each bank converts its diagram into a landscape vector, say a 4-dimensional vector sampled at $t = 0.2, 0.4, 0.6, 0.8$:

$$v_A = (0.10, 0.30, 0.30, 0.10) \quad v_B = (0.08, 0.29, 0.31, 0.12)$$

Step 3 — Hashing and collision probability.

The angle between v_A and v_B is small (the vectors point in nearly the same direction), roughly $\theta = 8$ degrees $= 0.14$ radians. The single-hyperplane collision probability is:

$$\Pr[\text{collision}] = 1 - \theta/\pi = 1 - 0.14/3.1416 \approx 1 - 0.045 = 0.955$$

Using several independent hyperplanes (say 20) and requiring most of them to agree pushes the combined false-positive rate down further, at the cost of also slightly raising the chance of missing a genuine match — a tunable trade-off, not a free win.

Step 4 — Sandboxed execution.

The collision resolves to a region id at each bank. A signed, distilled entity-linkage Wasm module (say, capped at 8 MB memory and 50 million compute steps) runs against only that region at each bank, and each returns a short result vector, at most 256 bytes, that a regulator combines to identify the shared entity — having received zero raw transaction records from either bank.

7. What's Actually True Today (The Honest Part)

As of this draft, the architecture above is fully specified, but none of the three components — Manifold-Core, TIP, or S-MCP — has a working, benchmarked implementation yet. That is the actual state of the project, not a caveat buried at the end.

OPEN, UNRESOLVED, STATED DIRECTLY

- Vietoris–Rips computation cost at production dataset scale is unmeasured.
- TIP's information leakage under repeated querying has no formal bound yet.
- Distillation accuracy loss for S-MCP modules versus their teacher model is unmeasured.
- LSH false-positive and false-negative rates need empirical characterization, not an assumption that they're negligible.

The roadmap is explicitly phase-gated so that each phase's measured result — including a result that doesn't support the plan as scoped — determines whether the next phase proceeds:

- **Phase 1 (Months 1–3):** build the manifold indexing pipeline; benchmark retrieval accuracy against standard cosine-similarity RAG on named, external multi-hop datasets.
- **Phase 2 (Months 3–5):** build the S-MCP sandbox and distillation pipeline; measure accuracy retention versus the teacher model; demonstrate zero data exfiltration.
- **Phase 3 (Months 5–8):** implement TIP over QUIC in full; run the two-bank worked example end-to-end across simulated nodes; commission an independent security review before any real pilot.

LIKE YOU'RE FIVE

Right now, ATLAS is a really good, really detailed drawing of a treehouse — with the measurements, the wood list, and the ladder plan all worked out. Nobody has actually built it and climbed up yet. That's the next step, and it's being done one board at a time, checking each board holds weight before adding the next.

This document is a plain-language and mathematical explainer draft based on the ATLAS technical README and roadmap materials. It is not the investment memorandum and makes no claims beyond what those source documents state.